

Total No. of Questions : 5]

SEAT No. :

PE-4200

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[6554]-2004

F.Y. M.C.A. (Management)

RMW-554 MJ : RESEARCH METHODOLOGY

(2024 Pattern) (Semester - II)

Time : 2½ Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) All questions carries equal marks.
- 2) Draw neat diagrams wherever necessary.
- 3) Scientific calculator is allowed.

Q1) a) Define Research Methodology and explain its significance in academic and professional contexts. [5]

b) Compare Qualitative and Quantitative research with suitable example. [5]

OR

c) Describe Research Process and Steps explain with diagram. [5]

d) Explain common research methods used in computer applications, providing relevant examples. [5]

Q2) a) Elaborate types of research with example. [5]

b) Explain Data Collection methods with suitable example. [5]

OR

c) Write note on Ethics in Research. [5]

d) Define research design with suitable diagram. [5]

Q3) a) What is sampling discuss the process of sampling.

Calculate the sample size for a population of 100000. Take confidence level as 95% and margin of error as 5% [10]

OR

b) Elaborate Probability sampling techniques with advantages and disadvantages. [10]

P.T.O.

Q4) a) Discuss Chi - square test.

From the data given below about treatment of 250 patients suffering from a diseases, state whether the new treatment is superior to the conventional treatment.

Treatment	Favourable	Non- Favourable	Total
New	140	30	170
Conventional	60	20	80
Total	200	50	250

For the above example Formulate null and alternate hypothesis and apply Chi-square test.

(consider: chi - squared value 0.05 significance level at degree of freedom 1 is = 3.84) **[10]**

OR

b) Two types of drugs were used on 6 and 5 patients for reducing their weight.

Drug A was imported & Drug B was Local. The decrease in weight (in kg) after using the drugs for 90 days was given below.

Drug A: 8,10,12,9,14,13

Drug B: 7,9, 14,12,8

Is there significant difference in effectiveness of Drug?

Apply two sample t - test for the above example (consider t value at 0.05 significance = 2.262). **[10]**

Q5) a) A MCA student is writing his research report on “Cybersecurity trends and threats”. He has completed his research but is unsure how to structure report. Based on the standard structure of a research report, suggest an appropriate structure for finalizing his research report. **[10]**

OR

b) Prepare an outline for a research on topic “Role of AI in Education “ with proper title , objectives , hypothesis , Research Gap , Data Collection sampling, Result and Discussion and Conclusion. **[10]**

